

QTagger+ — QSVM Track

Fidelity-kernel QSVM against classical baselines
for malware family attribution

2 August 2026

Abstract

QTagger+ compares classical machine-learning baselines against a fidelity-kernel quantum support-vector machine (QSVM) on malware classification, under a protocol designed so that the two are measured identically. It reports the QSVM track in full.

Two datasets are used, each with a binary and a 15-class task. On the binary task of CIC-MalMem every method exceeds 0.9998 macro-F1, so that task cannot rank methods and is treated as a control rather than a result. On the three configurations where the task discriminates, classical models beat the QSVM with $p < 0.05$ on all 84 paired comparisons. The QSVM’s failure on CIC 15-class is a property of the kernel rather than the data — a classical RBF kernel on the identical features does not concentrate, while the fidelity kernel collapses to $0.40\times$ its spread.

1 Classical baselines, and the saturation problem

The starting point is four classical models — random forest, XGBoost, LightGBM, and an SVM with an RBF kernel — tuned and evaluated on the full CIC-MalMem-2022 dataset: 58 596 rows, 55 memory-forensic features, 5-fold cross-validation with an Optuna inner loop.

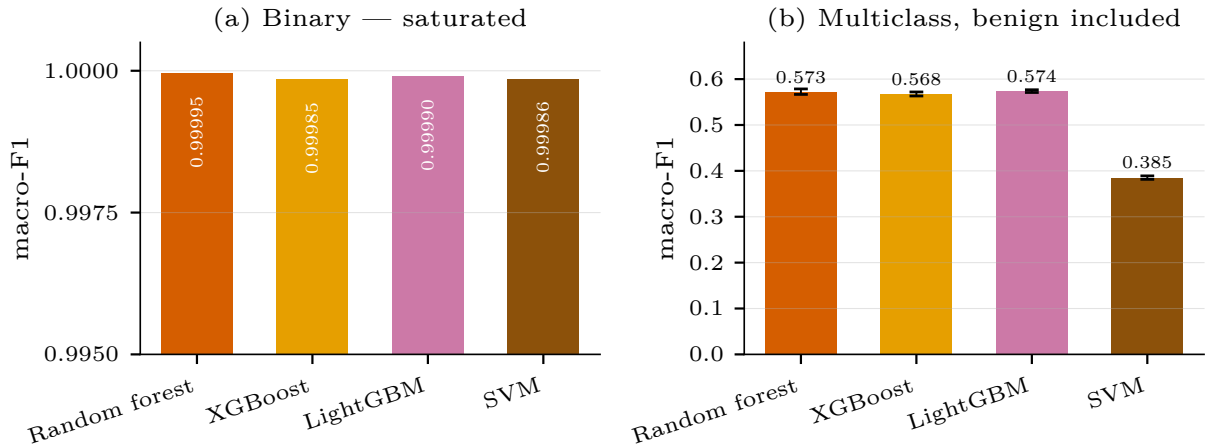


Figure 1: Full-dataset CIC-MalMem, no dimensionality reduction. (a) The binary task is saturated — all four models exceed 0.9998 macro-F1 and the spread between best and worst is 10^{-4} . (b) The multiclass task in its original form, benign still included as a class, separates the models by 0.19 macro-F1. These full-dataset runs predate experiment tracking in this project and are transcribed from its written record.

Figure 1(a) shapes everything that follows. When every model scores above 0.9998, the task cannot rank methods: any comparison on it measures noise. The project’s central claims are therefore made on the multiclass task, and a level result on CIC binary — which does occur — is read as a property of the task rather than evidence of parity.

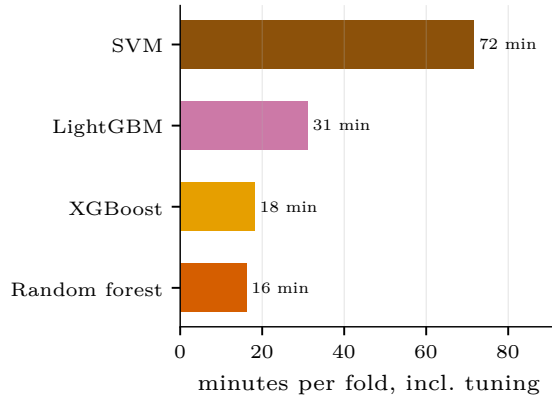


Figure 2: Tuning cost for a single outer fold on the full dataset. The SVM — the closest classical analogue to a QSVM, and therefore the most relevant comparison — is the most expensive at 72 minutes per fold.

Figure 2 sets a constraint that runs through the whole project: even the classical side is expensive at full scale, and the quantum side is quadratically worse. The subsampling design in §2.2 follows from it.

2 The comparison protocol

2.1 What makes the comparison fair

Both frameworks consume the output of one shared pipeline: a zero-variance filter, a correlation filter at 0.95, standardisation, then PCA to $n_{\text{components}}$ components. That single parameter fixes the classical feature count *and* the quantum qubit budget simultaneously. The pipeline is fit on training-fold rows only.

Because the fidelity kernel needs $O(n^2)$ circuit evaluations, quantum runs work on a stratified 1000-row subsample. The subsample and the five outer folds computed over it are written to disk by the quantum runner and *replayed* by the classical runner, so neither side chooses its own rows. A full-dataset classical run is never compared against a subsampled quantum run.

2.2 The cost that forces subsampling

Figure 3 is measured per circuit pair rather than per fold, which removes the effect of subsample size changing across the project. At $n = 1000$ a training fold needs roughly 320 000 pair evaluations, so the per-pair figures translate to minutes per Gram at the low end and tens of minutes at $n_{\text{qubits}} = 8$.

A sizing exercise fitted this scaling law and solved it for a target budget, recommending $n = 900$ at 60s per Gram build, with a class-balance floor of $n \geq 750$ for the 15-class task. The 1000-row subsample used throughout sits just above that floor.

2.3 Binary is a ceiling, on the subsample too

Figure 4 shows that the saturation of Figure 1(a) survives dimensionality reduction and subsampling. The QSVM does sit level with the classical models here. It sits level with them because nothing separates on this task, not because the quantum kernel is competitive.

2.4 The multiclass reframe

The original multiclass task kept benign as a 16th class. Since benign detection is exactly what the binary task measures, keeping it made the label space pathologically imbalanced and

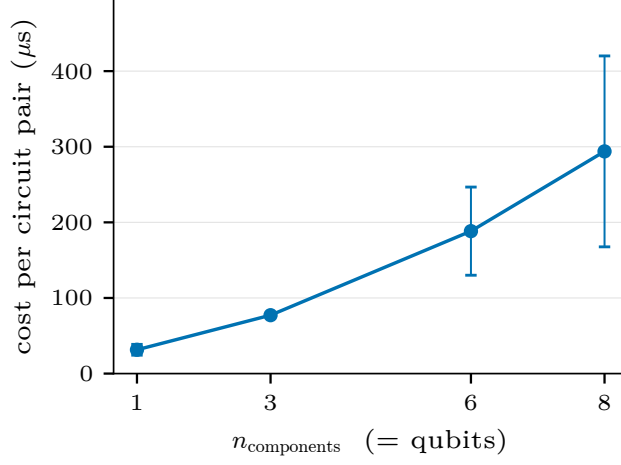


Figure 3: Per-circuit-pair cost of building the training Gram matrix, against qubit count, from a single session so the worker configuration is constant. Cost per pair rises roughly tenfold from 1 to 8 qubits — and the number of pairs itself grows as n^2 , so the two effects compound.

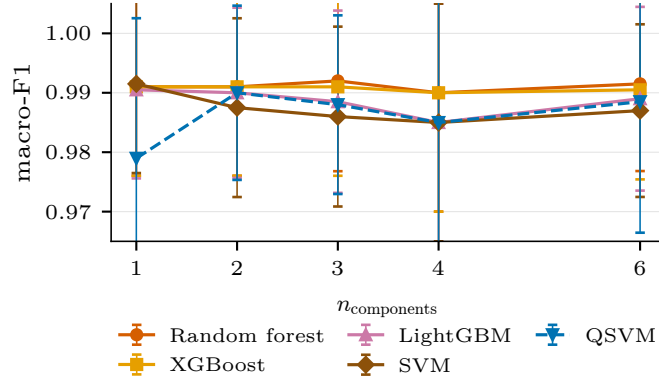


Figure 4: CIC-MalMem binary on the 1000-row subsample, across the PCA sweep. Every model, quantum and classical, lies between 0.977 and 0.992 macro-F1, and every difference is inside the fold-to-fold error bars. Aggregated over all recorded folds: these runs predate parent/child run tracking, so individual sweeps are not recoverable (§7).

duplicated the binary task inside the multiclass one. The task was reframed: benign rows are dropped, leaving 29 298 malware rows across 15 families. All 15-class numbers in this report use that definition, which is why they are not comparable to Figure 1(b).

3 A second dataset, and the encoding comparison

3.1 EMBER 2018

A second dataset tests whether any conclusion is an artifact of CIC. EMBER 2018 contributes static PE features in 2381 dimensions. Work uses an 18 014-row extract of the published test set; the full set is a single $200\,000 \times 2384$ block that cannot be read incrementally. Its 15-class task keeps the 15 largest families and downsamples each to the smallest, giving 14 325 rows at exactly 955 per family — a balanced counterpart to CIC’s $1.7\times$ imbalance.

3.2 angle versus IQP

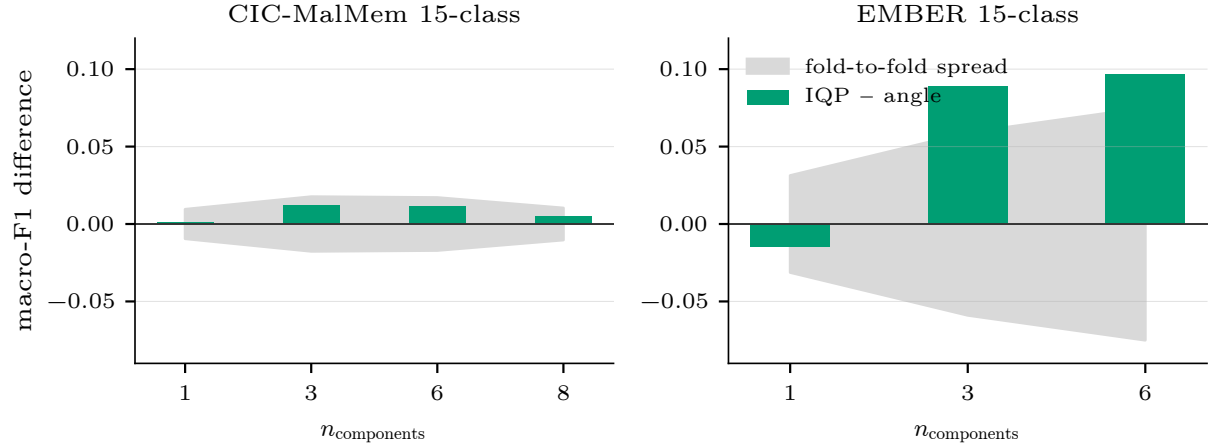


Figure 5: IQP minus angle encoding, 15-class macro-F1. The grey band is the fold-to-fold spread of the encodings themselves. On CIC every difference lies inside that band. On EMBER the differences at $n_{\text{components}} = 3$ and 6 clear it.

Figure 5 gives the verdict: IQP is never worse than angle, but the advantage is only measurable on EMBER. On CIC the deltas are $+0.0009$, $+0.0122$, $+0.0113$ and $+0.0046$ at $n_{\text{components}} = 1, 3, 6, 8$, each smaller than either encoding’s own standard deviation. Extending to 8 qubits did not rescue CIC 15-class performance (0.0739, against a 0.067 chance baseline), which ruled out qubit count as the missing ingredient.

A methodological change came out of this: comparing encodings by counting which one an inner-CV tuner selected per fold was retired. It conflates “which was picked when both were live options” with “which performs better”, and produced an ambiguous 8/15 win count on data where the direct macro-F1 deltas showed IQP ahead at every $n_{\text{components}}$. Encoding comparisons now use dedicated single-encoding runs and report deltas with their spread.

3.3 SOREL-20M

A third dataset was scoped and its multiclass labelling scheme designed and validated over 19.7M rows: dominant-tag argmax over 11 behaviour-tag counts, ties broken by declared column order, all-zero-tag rows dropped. The feature store was never fetched and no sweep was run — these are independent pieces of work and only the first was completed. §6 states the blocker.

4 Results

4.1 Classical against quantum on EMBER

An early version of this comparison ran SVM only on the classical side, which measured the QSVM against a sub-optimal baseline. The full four-model table:

Table 1: EMBER 2018, held-out macro-F1, mean over 5 outer folds. Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_{\text{components}} = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_{\text{components}} = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_{\text{components}} = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_{\text{components}} = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_{\text{components}} = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_{\text{components}} = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

Random forest beats SVM on every binary cell, which widens the classical advantage from $+0.0727$ to $+0.0995$ at $n_{\text{components}} = 1$, from $+0.1222$ to $+0.1491$ at 3, and from $+0.1587$ to $+0.1697$ at 6. On the 15-class task SVM is genuinely the strongest baseline at $n_{\text{components}} \in \{3, 6\}$. Four of six cells change, and every one changes against the quantum pipeline.

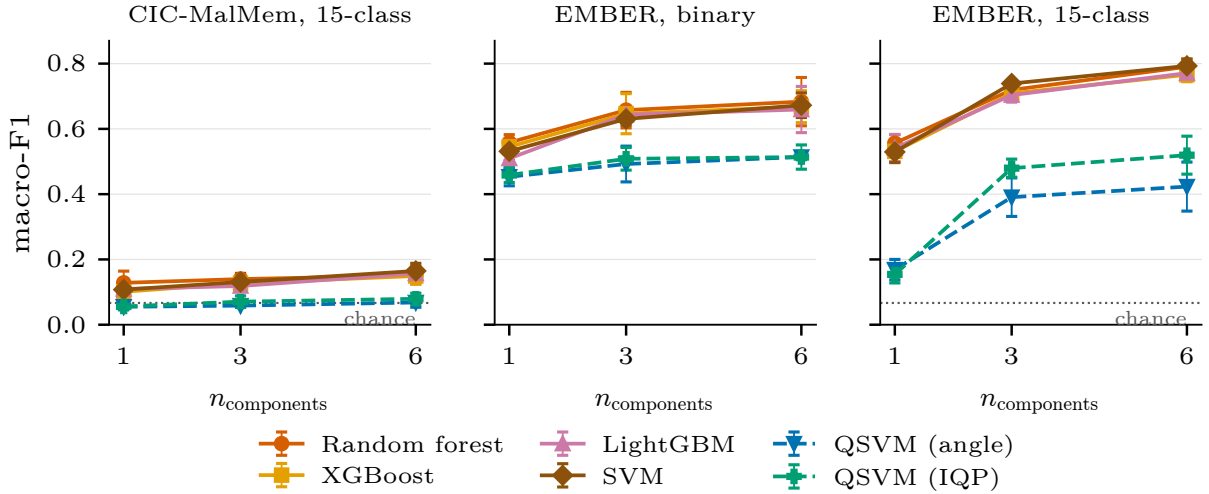


Figure 6: Macro-F1 against the shared feature/qubit budget. Solid lines are classical, dashed the QSVM encodings. On EMBER binary the QSVM gains 0.055 macro-F1 from one to six components where the random forest gains 0.125: the gap widens with capacity rather than closing.

4.2 The subsample the project rests on

Every quantum result is computed on 1000 rows, so the subsample must represent its population. Read against the null expectation — a faithful subsample still rejects about $\alpha \cdot n_{\text{features}}$ features by chance — CIC rejects 0 of 55 where ≈ 2.75 were expected, and EMBER 1 of 2381 where ≈ 119 were. Maximum class-proportion drift is below 0.001 on both. Both subsamples are faithful.

4.3 Why CIC 15-class collapses

The QSVM sits at the random baseline on CIC 15-class (0.056–0.079 against chance ≈ 0.067) while reaching 0.42–0.52 on EMBER 15-class. Class imbalance and qubit count are both ruled out — EMBER’s families are exactly balanced and CIC’s only $1.7\times$ imbalanced, and extending CIC to 8 qubits does not help. Two further measurements separate a dataset effect from a kernel effect.

Feature geometry — a dataset effect. On the identical projected features, EMBER’s families are $\approx 8\times$ more separable by Fisher ratio and up to $50\times$ on the worst-case family. CIC’s silhouette is negative at every $n_{\text{components}}$: the average point lies closer to another family’s centroid than its own. CIC’s PCA nonetheless explains *more* variance (33–81% against 8–30%), so its variance sits in directions that do not discriminate family. This explains why every method does badly on CIC 15-class, and cannot explain the QSVM-specific failure — a classical SVM extracts 0.108–0.165 from these same features where the QSVM extracts 0.056–0.079.

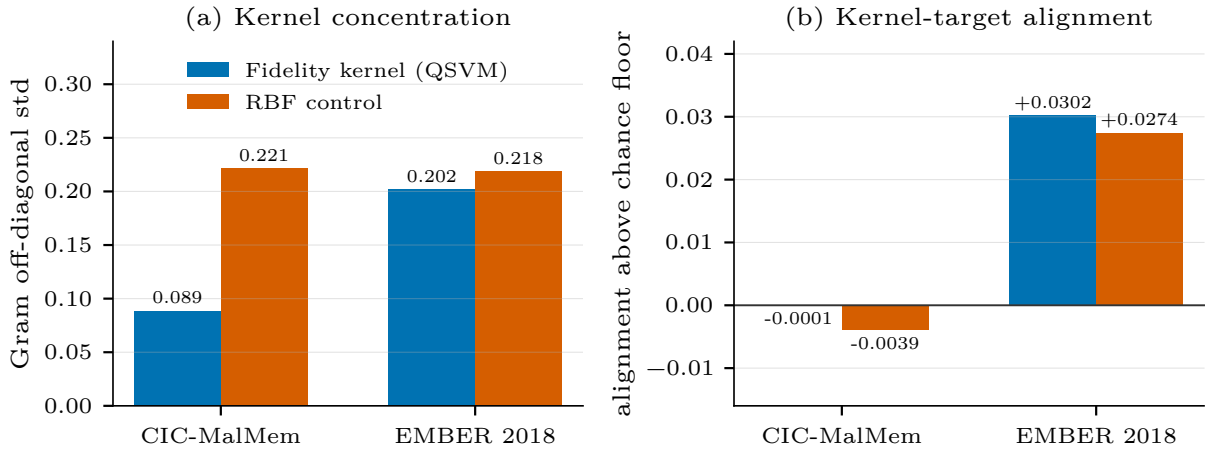


Figure 7: First outer training fold, $n_{\text{components}} = 6$, IQP encoding, 800 rows, with an RBF control built on the *identical* feature matrix. (a) Off-diagonal Gram spread. (b) Kernel-target alignment relative to its floor — a constant kernel scores $1/\sqrt{15} \approx 0.258$ on 15 balanced classes, so only the excess carries information.

Kernel geometry — a kernel effect. The RBF control produces near-identical spread on both datasets (0.2212 against 0.2184), so CIC’s features do not make a classical kernel concentrate. The fidelity kernel tracks RBF on EMBER (0.92 $\times$) and collapses to **0.40 $\times$** on CIC: every off-diagonal similarity squeezed into a band $2.5\times$ narrower from the same inputs. The Gram approaches a scaled identity, and a support-vector machine given that has nothing to separate on.

On alignment, the CIC fidelity kernel lands 0.000069 *below* the constant-kernel floor — indistinguishable from an all-ones matrix — and the RBF control is below it too. On EMBER both clear it by ≈ 0.03 . So alignment separates the datasets and concentration separates the kernels.

Two limits on this. First, **bandwidth** is the named mechanism for concentration and has never been tuned on CIC beyond its default; no sweep was run, so this is a hypothesis with an identified lever rather than a measured fix. Second, alignment reports that neither kernel carries label information on CIC, yet a classical SVM reaches 0.16 there — alignment is a global unweighted average over all pairs and an SVM requires no such thing, which makes it a poor predictor of achievable SVM performance.

4.4 Significance

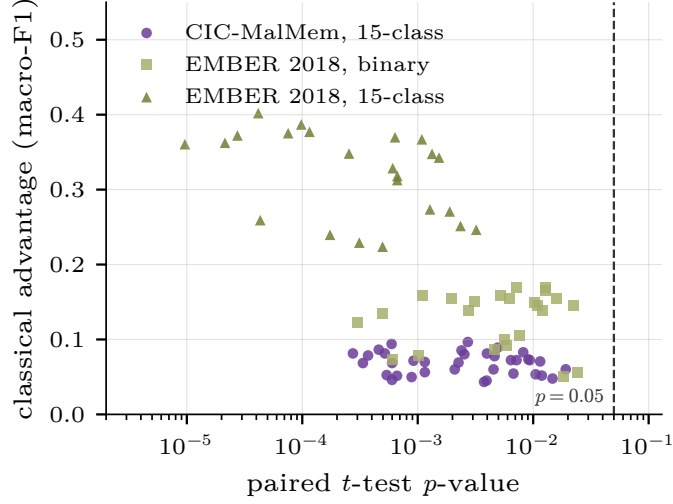


Figure 8: All 84 testable quantum–classical pairs. Every point lies left of $p = 0.05$ and above zero.

All 84 testable pairs favour the classical model and all 84 reach $p < 0.05$ on the paired t -test; the worst p -value across every pair is 0.0241.

The Wilcoxon signed-rank test returned exactly 0.0625 for all 84 pairs. With five paired samples that is the smallest attainable two-sided p -value, so the test cannot reach 0.05 at this fold count regardless of effect size; all 84 hitting the floor means every pair achieved the most extreme rank configuration available. Six folds would put 0.03125 within reach.

5 The comparison in one chart

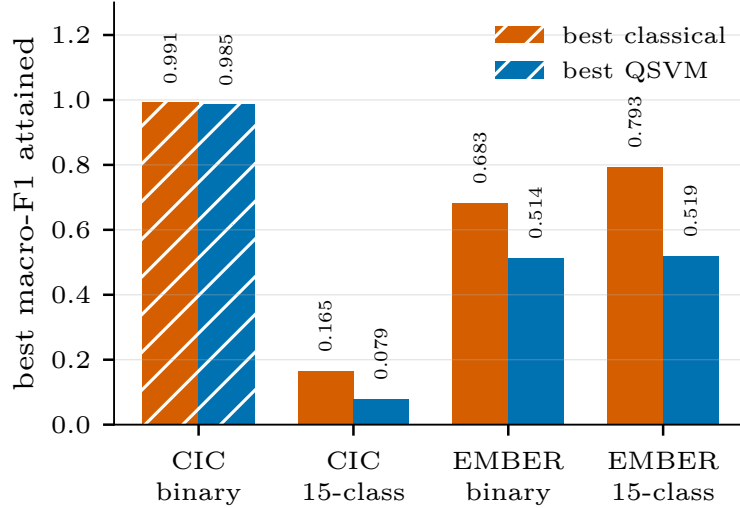


Figure 9: Best classical against best QSVM in each dataset/task cell, over $n_{\text{components}} \in \{1, 3, 6\}$. Hatched bars are aggregated from raw folds because no sweep boundary is recoverable for that cell (§7); the others come from the most recent complete 5-fold sweep.

Figure 9 is the project in one chart. On CIC binary the two are 0.006 macro-F1 apart — a tie at the ceiling, where nothing separates. In the three cells where the task can discriminate, the classical model leads by 0.085 (CIC 15-class), 0.170 (EMBER binary) and 0.274 (EMBER 15-class).

6 What was not done

SOREL-20M sweeps

Closed rather than deferred. Neither disk (227 GiB free against 71.6 GiB needed) nor memory is the blocker — the feature store is a memory-mapped LMDB whose reads page through the OS cache. The blocker is the fixed 71.6 GiB transfer: the store offers no key-level remote access, so there is no way to fetch only the thousand rows a sweep would need. Resume conditions are documented.

Variational quantum classifier

Not implemented. The comparison is fidelity-kernel QSVM against classical baselines, not “quantum methods” against classical ones.

Basis encoding

Deliberately omitted. It maps discrete values onto computational basis states, but the pipeline’s inputs are continuous PCA components, so it would require a quantisation step that discards what PCA was used to retain.

UMAP as a second reduction

Deliberately omitted. PCA is the classical/quantum alignment point; a second reduction with no classical counterpart would break it.

CTGAN-augmented run

Blocked on the augmented dataset from another team. The pipeline accepts any parquet, so nothing is outstanding on this side.

Weighted F1

Implemented, but no sweep was re-run afterwards, so no result in this report carries it. It matters most on CIC 15-class, where families run 48–82 rows in the subsample.

7 Untestable configurations

All CIC binary comparisons — 9 cells across 4 models — cannot be significance-tested. Every CIC binary run, classical and quantum alike, predates parent/child run tracking in the experiment log and carries no recoverable sweep boundary; the 45 fold records per model resolve into two distinct sweeps per $n_{\text{components}}$ with different configurations and no identifier separating them. The records are valid measurements and are used descriptively in Figures 4 and 9; no grouping was reconstructed, since the time gaps between runs are not a recorded boundary. The limitation is symmetric across frameworks.

McNemar’s test between frameworks needs per-sample predictions from both models on a shared test set. QSVM predictions were not persisted during these sweeps; persistence was added afterwards but no sweep was re-run.

8 Method notes

- **Macro-F1, not accuracy.** On EMBER binary at $n_{\text{components}} = 1$ the QSVM attains 0.890 accuracy in three of five folds while the benign class’s F1 is exactly 0.000 in those same folds, because 88.9% of rows are malware and the model labels everything malware.
- **Runs are grouped by sweep, not by parameters.** The same configuration was swept several times across the project, so a parameter-based grouping returns 10, 15 or 20 records for a five-fold sweep and averages distinct hyperparameter searches together — 57 parameter groups in the experiment log have a size other than five. For the QSVM, encoding is additionally a *per-fold outcome* rather than a configuration key: the two-tier tuner selects it inside each fold, so one joint sweep can appear as four angle folds and

one IQP fold. Runs are grouped by parent run, keeping only sweeps that completed with exactly five folds. 123 of 125 nested sweeps are clean.

- **Simulator only.** All results use a state-vector simulator. No hardware execution and no noise model.

9 Summary

Across two datasets and two tasks, classical models beat the fidelity-kernel QSVM on every configuration where the task can discriminate at all, with $p < 0.05$ on all 84 paired comparisons. The single configuration where the two were level — CIC binary — is saturated for every method, and level results there carry no information about the methods.

The transferable finding is the mechanism rather than the ranking. The fidelity kernel does not fail on CIC because CIC’s features are hard: a classical RBF kernel on identical features does not concentrate. It fails because the fidelity kernel itself collapses to $0.40\times$ the classical kernel’s spread on that data, driving its Gram matrix toward a scaled identity. That is a statement about the method, and it predicts where the approach will struggle on data nobody in this project has seen.

Next steps. Tune **bandwidth** on CIC, the identified mechanism whose lever remains untested. Move to six outer folds so a non-parametric test becomes usable.